Lukas Liakhovitch

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EDUCATION

Purdue University, *B.S. in Mechanical Engineering* (Aug 2025 - Dec 2028), West Lafayette, IN
GPA 3.55/4, Dean's List '25, Semester Honors '25 & '26, Alpha Lambda Delta, Phi Eta Sigma Honor Societies
Relevant Coursework: Multi-variate & Vector Calculus, Differential Equations, Statics, Dynamics, Thermodynamics

SKILLS

TECHNICAL: Java, C, MATLAB, Python, Claude Code, Linux/WSL, Fusion360, Onshape, NX/SimLab, machining, fabrication
SOFT: Project Management, Team Leadership, Communications, Advocacy, Problem-solving, Root-cause Analysis

EXPERIENCE

Mechanisms Engineer (August 2026 - Present)
***Purdue Aerial Robotics Team**, West Lafayette, IN*
- Engineered all control surfaces (ailerons, elevators, rudder) for the 2026-27 SUAS aircraft in Onshape; sized 3 servo-controlled drive systems for minimum weight, then built and tested 3D-printed prototypes
- Integrated fully parametric Onshape models into a 30+ contributor, 4-subteam master assembly, managing branching, merging, and version control to ensure cross-team CAD compatibility and reliable design integration

Software Engineering Intern (May 2026 - Present)
***Eukodyne**, Princeton, NJ*
- Redesigned the company website; developed interactive product demos illustrating software and product features, directly driving the company's acceptance into the Microsoft AI Cloud Partner Program
- Ported legacy software to a cross-platform React progressive Web App for mobile; deployed an MCP-based natural language chatbot within the Web app to orchestrate database workflows and system-level actions
- Built Quarry, a safety-gated automation pipeline that turns web sources into validated, fully cited JSON process templates; developed auto code review loops using an AI-assisted DLC (Claude Code, GitHub Copilot)

Active Controls Structures and Manufacturing Engineer (January 2026 - May 2026)
***Purdue Space Program**, West Lafayette, IN*
- Modeled and manufactured structural members and fluid system mounts for a 9 ft, 500 lb self-landing rocket in NX
- Performed FEA in NX SimLab and ANSYS on lander components, iterating designs to meet structural requirements

Software Engineering Intern (May 2025 - May 2026)
***Clearcut Software**, Fair Lawn, NJ*
- Deployed financial processing library (Pyxirr) from Rust to C++ and created custom DuckDB extension for its use
- Developed SQL and Python unit tests to validate extension functionality and format results for presentation

Torch Igniter Rocket Engine & Turbopump Development Team Member (September 2025 - January 2026)
***Purdue Undergraduate Rocket Propulsion Lab,** West Lafayette, IN*
- Designed and fabricated central shaft bearing housing for first undergrad student-developed turbopump
- CAD-ed custom sized and configured torch igniter rocket engine (team of 4)

Founder & Chief Engineer (April 2023 - July 2025); **Mentor** (Aug 2025 - Present)
***FIRST Tech Challenge (FTC) Robotics - Ultraviolet 23268**, Hackettstown, NJ*
- Founded, recruited and managed new independent, award-winning team; ongoing mentor & advisor
- Designed and built 18x18 in robot in Fusion 360; implemented custom pathfinding and control algorithms (PID, motion profiling, inverse kinematics) in Java to optimize autonomous navigation, focusing on consistency
- Won NJ State Championship (190 teams) and advanced to FIRST World Championship (in only 2nd season); achieved the 2nd highest average score world-wide and ranked #4 (out of 8,000+ teams) globally

PROJECTS

Restoration and Repair (May 2024 - Aug 2025)
- Used vehicle restoration: diagnosed issues, replaced worn or missing parts (exhaust, radiator, brakes, & audio system)
- Laser cutter repair: exchanged the command module, fixed localization, control, and software issues